



APT 3020B – Knowledge Representation Practical Lab

Building a Simple Medical Expert System Using Knowledge Representation

OBJECTIVE

By the end of this lab, students should be able to:

- Represent knowledge using facts and rules
- Design a rule-based expert system
- Apply logical inference
- Organize and document a project using GitHub

PRACTICAL SCENARIO

A hospital is developing a simple intelligent medical expert system to identify possible illnesses based on symptoms.

The system should use the following symptoms:

1. Fever
2. Headache
3. Cough
4. Chest Pain
5. Sneezing
6. Runny Nose
7. Fatigue
8. Sore Throat
9. Vomiting
10. Diarrhea

The system should support the following rules:

- IF Fever AND Headache AND Fatigue → Malaria
- IF Cough AND Chest Pain AND Fatigue → Pneumonia
- IF Sneezing AND Runny Nose AND Sore Throat → Flu
- IF Vomiting AND Diarrhea AND Fatigue → Food Poisoning

TASKS

TASK 1: KNOWLEDGE BASE DESIGN (4 MARKS)

Create a knowledge base containing:

- Facts
- Symptoms
- Diseases
- Rules

The knowledge base can be stored in:

- Python dictionaries
- JSON
- Text files
- Lists or arrays

TASK 2: RULE-BASED SYSTEM (6 MARKS)

Develop a simple rule-based system that:

- Accepts symptoms as input
- Applies IF–THEN rules
- Displays the possible illness

The system should correctly infer diseases based on the rules provided.

TASK 3: SEMANTIC NETWORK DIAGRAM (4 MARKS)

Create a semantic network diagram showing relationships between:

- Patients
- Symptoms
- Diseases

You may use:

- Draw.io
- PowerPoint
- Lucidchart
- Hand-drawn diagram (scanned clearly)

Save the diagram inside the repository.

TASK 4: DOCUMENTATION AND PROJECT STRUCTURE (6 MARKS)

Your GitHub repository must contain:

Required Files

- README.md
- Source code
- Semantic network diagram
- Sample outputs/screenshots

README REQUIREMENTS

Your README.md should include:

1. Project Title
2. Project Description
3. Objectives
4. Symptoms Used
5. Diseases Detected
6. Rules Applied
7. Technologies Used
8. How to Run the Program
9. Sample Output Screenshots
10. Group Members (if applicable)

REPOSITORY STRUCTURE EXAMPLE

APT3020-Knowledge-Representation-Lab/

```
|
|— README.md
|— main.py
|— knowledge_base.json
|— semantic_network.png
|— screenshots/
|   |— output.png
|— docs/
|   |— report.pdf
```

SUBMISSION INSTRUCTIONS

1. Push your completed project to GitHub.
2. Ensure the repository is public.
3. Submit ONLY the GitHub repository link on Blackboard.
4. Poorly documented repositories will lose marks.
5. Repositories without a README file will NOT be marked.

BONUS ACTIVITY (+2 MARKS)

Add:

- User input validation

- More diseases and symptoms
- GUI or web interface
- Voice-based interaction
- AI chatbot integration